



< The Washington Institute for STEM, Entrepreneurship and Research >

Preparing for Quantum Hardware

Finding the right way into Quantum Integration.

PREPARED FOR

Vanguard

PREPARED BY

Elijah Kim

DATE

August 2026

Executive summary

There is no universally best quantum computer. There is a best machine for a particular workload, and it changes every eighteen months.

01 Every Quantum hardware vendor is telling the truth

Each quotes the one metric its architecture wins on.

02 Modality follows application

Choosing the machine first is the costly mistake.

03 Access beats ownership

Benchmark six vendors for less than one procurement cycle. No lock-in to an architecture that may not win.

WHAT WE RECOMMEND

- Pick one workload
- Benchmark, don't buy
- Score on seven axes
- Treat post-quantum cryptography separately
- Re-score twice a year

Scope of this briefing

Two halves: what the hardware is, and how to choose between the companies selling it.

● In scope

- The seven qubit modalities, and how each is built
- The vendors, and what they can deliver today
- Which algorithms map to which hardware
- A decision tree from problem to shortlist
- The software and access layers in between

● Out of scope today

- A single machine recommendation. WISER is platform-agnostic
- Procurement pricing, which is negotiated and moves fast
- Quantum sensing, networking and QKD, separate tracks
- The full post-quantum migration plan, flagged as its own programme
- Vendor-specific implementation, which follows the shortlist

< why every Quantum vendor sounds like the winner >

Six ways a benchmark can be true and still mislead

None of these claims are false. Each measures a different thing, and every vendor leads on the one it wins.

01

Physical qubit count

The headline. Says nothing about connectivity or error rate. A 4,400-qubit annealer and a 98-qubit ion trap can't run the same programs.

02

Logical qubit count

Error-detected and error-corrected differ by a factor of two. Helios reports 94 of one, 48 of the other, from the same 98 qubits.

03

Gate fidelity

Best pair or median. Prototype or product. IonQ's record 99.9923% is a prototype; its commercial systems quote 99.6%.

04

Vendor-defined benchmarks

#AQ is IonQ's. CLOPS and Quantum Volume are IBM's. Each was designed where its author is strong.

05

Quantum advantage claims

The task suits the hardware, and classical algorithms catch up. Google's 2019 claim was contested within days.

06

Roadmap dates

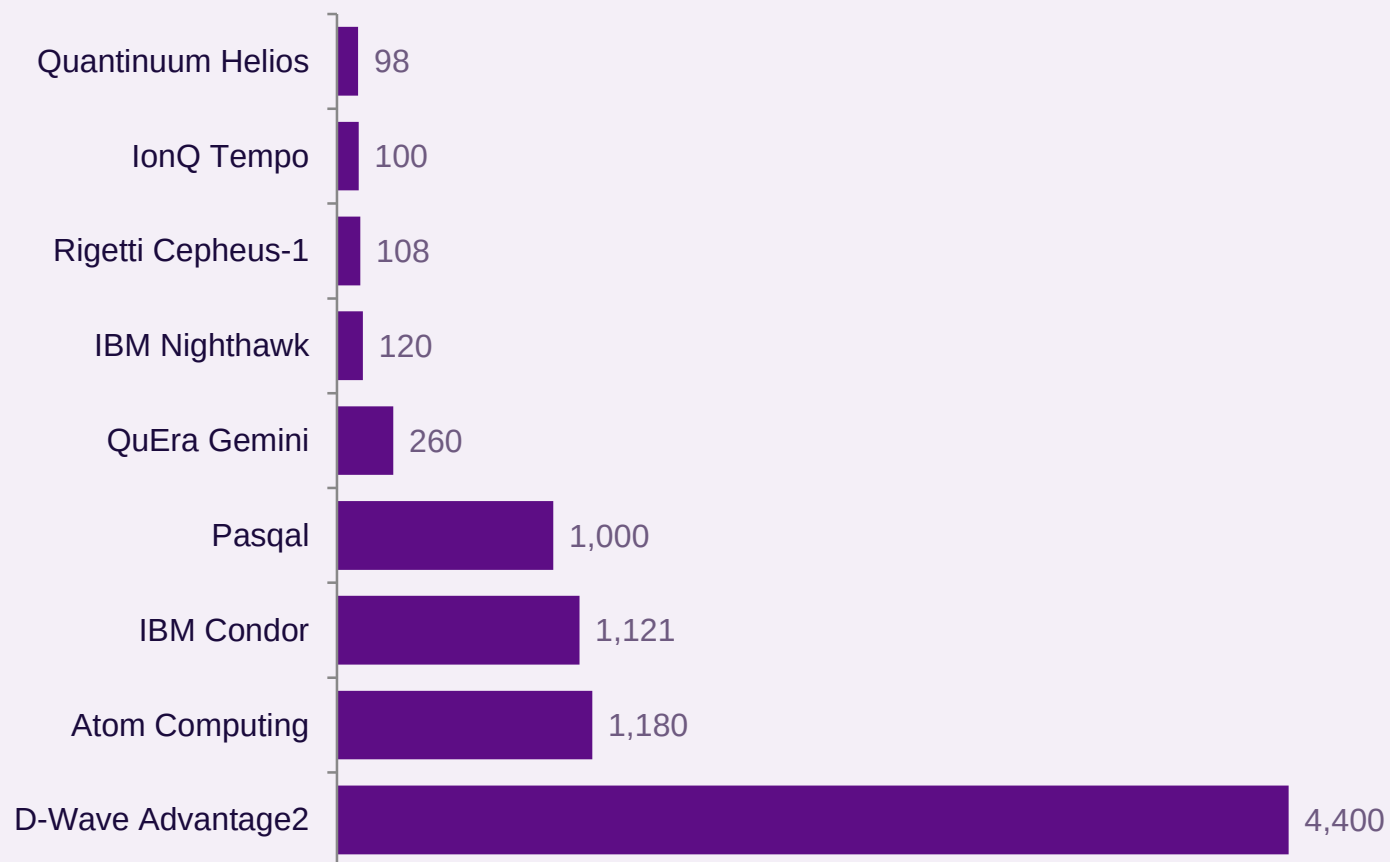
Ambitions, not commitments. Microsoft halved its own fault-tolerance timeline in 2026. The delivery record is mixed.

Same machines. Two different rankings.

The same systems, ranked by the press-release number, then by the number that decides whether an algorithm finishes.

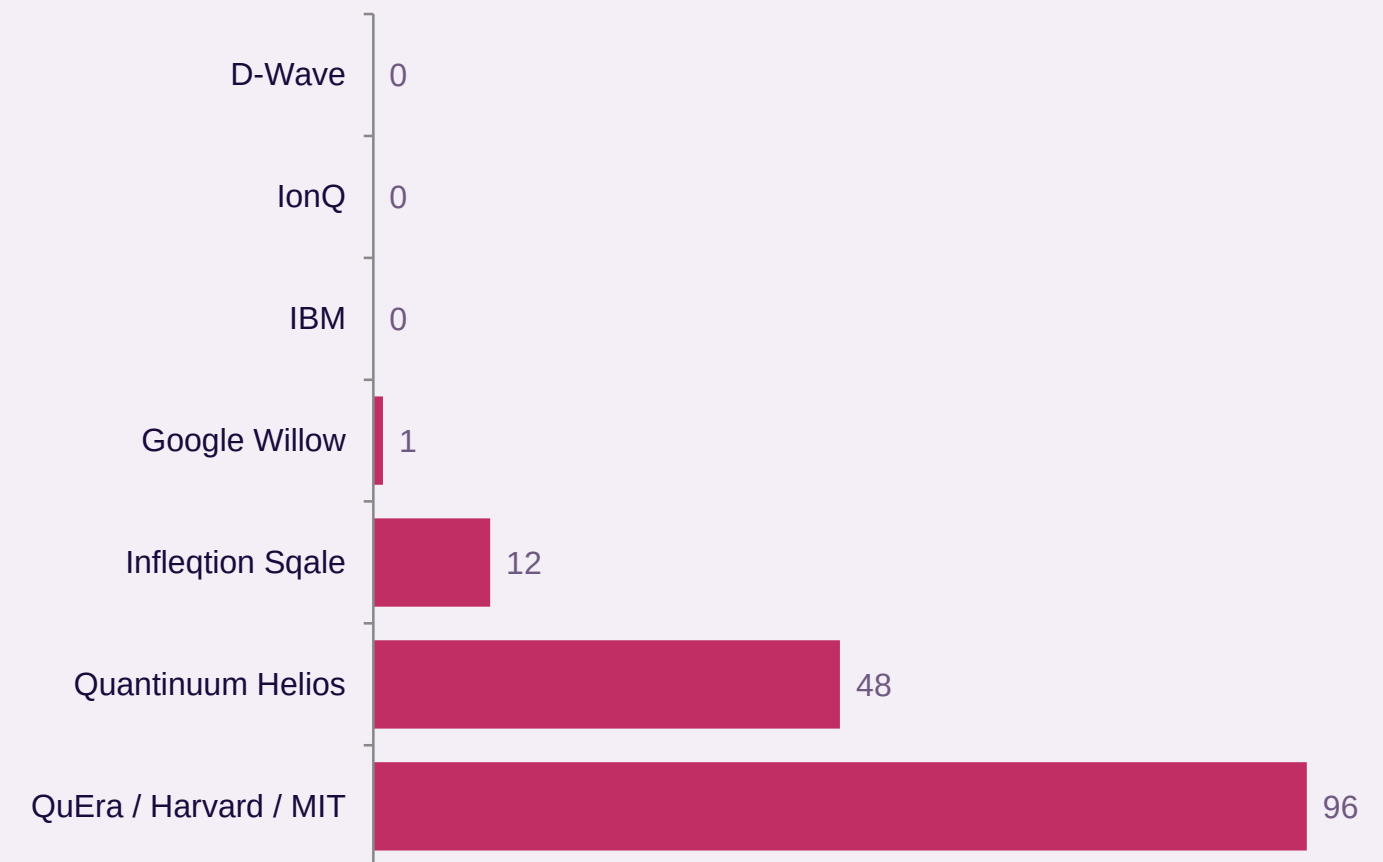
Ranked by physical qubits

The marketing number



Ranked by logical qubits

The number that decides whether an algorithm finishes



Physical counts: vendor figures, mid-2026. Logical counts: peer-reviewed, error-corrected only - QuEra 96 from 448 atoms (Nature, Jan 2026), Quantinuum 48 from 98 ions (Nov 2025), Infleqtion 12, Google 1 on Willow. IBM, IonQ and Microsoft publish targets, not verified counts. D-Wave's annealer has no logical-qubit axis.

Qubit Modality Review

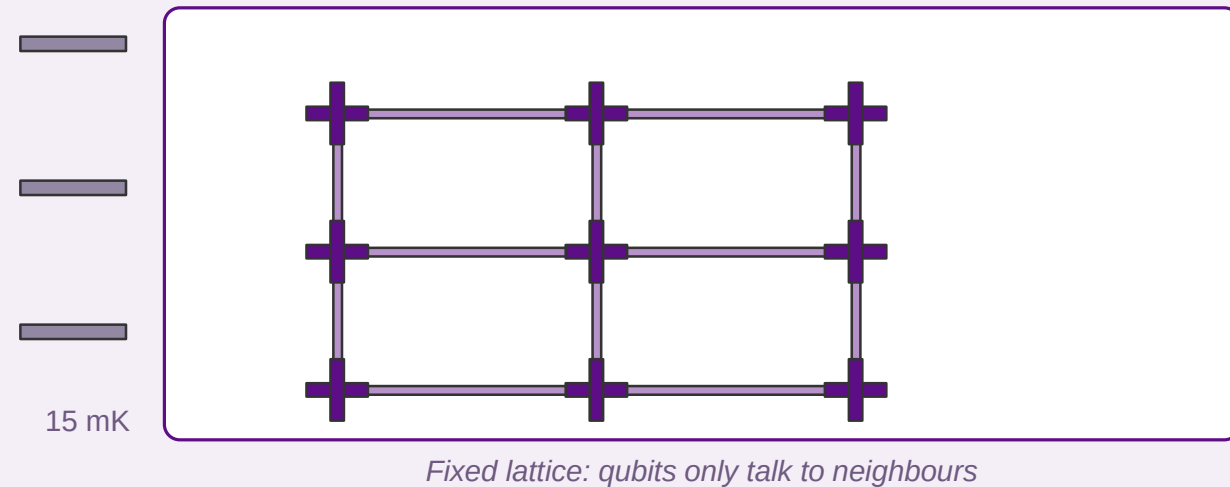
Every quantum company builds one of these seven.

PLATFORM	HOW IT WORKS	WHO BUILDS IT	WHY IT MATTERS
 Superconducting	Microwave circuits on a chip, cooled to 15 millikelvin	IBM · Google · Rigetti · IQM · AWS	<i>Most mature gate-model platform; fastest gates</i>
 Trapped ion	Individual charged atoms held in electromagnetic traps	Quantinuum · IonQ · AQT · Oxford Ionics	<i>Highest fidelities and all-to-all connectivity</i>
 Neutral atom	Uncharged atoms held in arrays of laser tweezers	QuEra · Pasqal · Atom Computing · Infleqtion	<i>Currently leads on verified logical qubits</i>
 Photonic	Single photons or squeezed light in waveguides	PsiQuantum · Xanadu · Quandela · ORCA	<i>Room temperature; native networking</i>
 Silicon spin	Single electron spins in CMOS quantum dots	Diraq · SQC · Quantum Motion · Intel · Equal1	<i>Smallest footprint; existing 300mm fabs</i>
 Topological	Information stored in Majorana zero modes	Microsoft	<i>Highest ceiling, earliest stage, highest risk</i>
 Quantum annealing	A tunable Ising machine, not a universal computer	D-Wave	<i>The only platform in routine production use</i>

Superconducting qubits

Superconducting qubits are essentially tiny engineered circuits operating at extremely low temperatures.

PHYSICAL STRUCTURE



120

qubits on IBM Nighthawk, with 218 tunable couplers

~1 ms

coherence time, the shortest of any modality here

5,000

two-qubit gates per job on current Nighthawk hardware

● Where it wins

- Maturity
- Speed.

● Where it struggles

- Scaling
- Coherence
- Wiring

How it is built

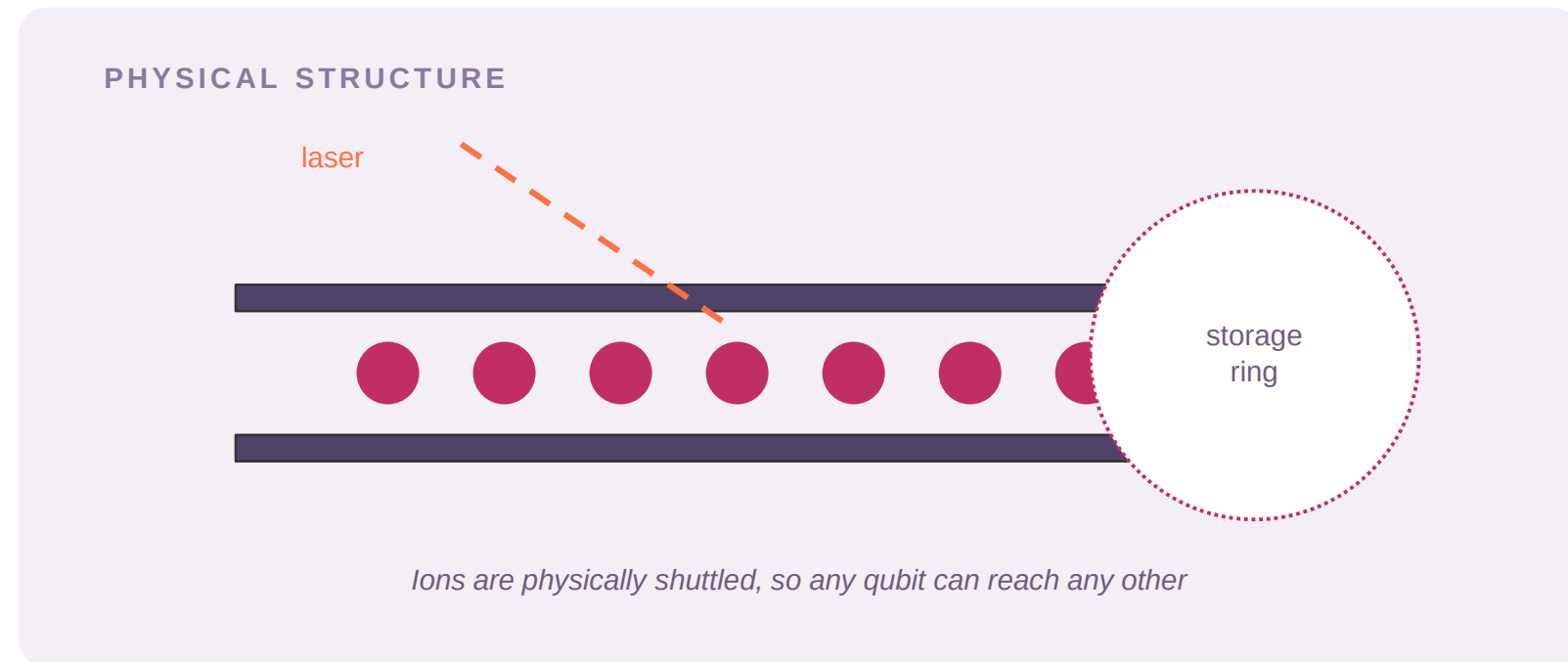
Superconducting circuits are patterned on a silicon wafer like a classical chip, then cooled to 15 millikelvin in a dilution refrigerator. Each qubit needs its own coaxial control line into the fridge. That wiring, not the qubits, is the scaling bottleneck.

WHO BUILDS IT

IBM · Google Quantum AI · Rigetti · IQM · AWS (cat qubits) · SEEQC · D-Wave (gate-model, via Quantum Circuits)

Trapped ion qubits

Individual charged atoms are held in electromagnetic traps. Two internal energy levels form the qubit; lasers do everything from preparation to readout.



How it is built

Charged atoms are held above a trap chip by radio-frequency fields in ultra-high vacuum. Every ion is identical, so there is no fabrication variability. Quantinuum shuttles ions between a storage ring and gate zones, giving true all-to-all connectivity.

WHO BUILDS IT

Quantinuum · IonQ · Alpine Quantum Technologies · Oxford Ionics (IonQ) · eleQtron

48

error-corrected logical qubits
shipping on Quantinuum Helios

99.99%

best two-qubit gate fidelity, an
IonQ prototype result

>10 min

coherence time, the longest of
any platform

● Where it wins

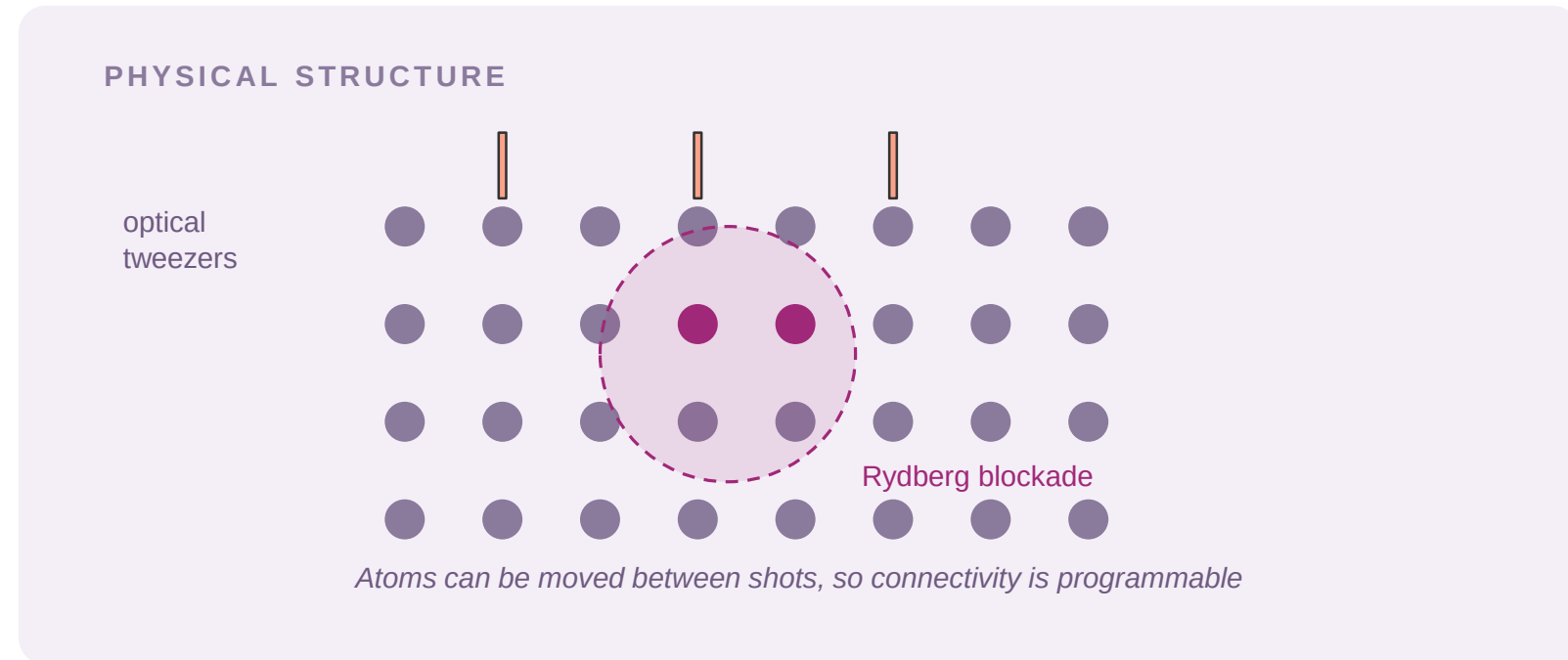
- Fidelity
- Connectivity
- Coherence

● Where it struggles

- Gate speed
- Physical scale
- Engineering complexity

Neutral atom qubits

Uncharged atoms held in a reconfigurable grid of laser beams.



How it is built

Atoms are laser-cooled and held in optical tweezers, arranged in software so the lattice can be rearranged between shots. Gates fire by exciting neighbours into Rydberg states. Atoms move mid-computation, which is why this platform holds the recent error-correction records.

WHO BUILDS IT

QuEra · Pasqal · Atom Computing · Infleqion · planqc · Google (Boulder, from 2026)

96

logical qubits from 448 atoms, the current verified record

1,180

atoms trapped in a single Atom Computing array

~3 kW

power draw of a rack-mounted Pasqal Orion at room temperature

Where it wins

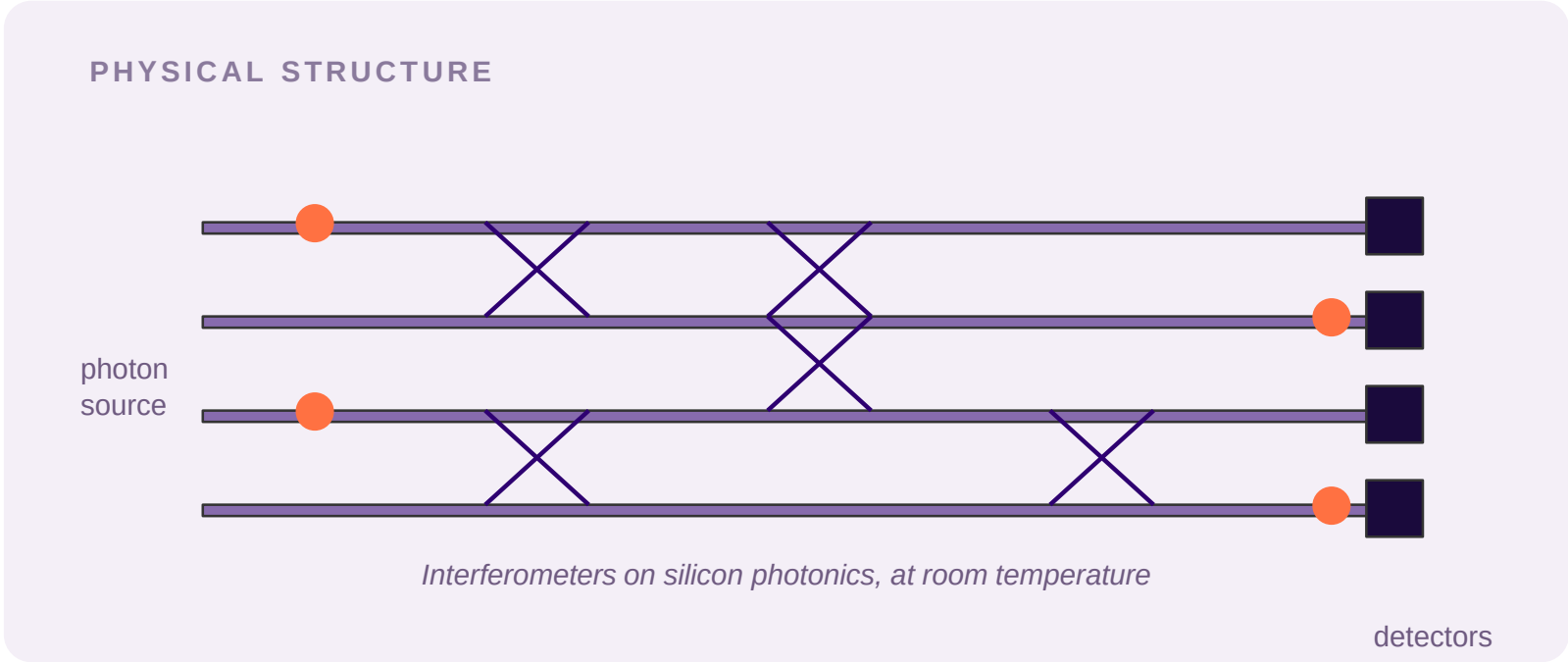
- Large physical scale
- Flexible connectivity
- Promising error correction.

Where it struggles

- Optical control
- Atom loss remain challenges.

Photonic qubits

Information is carried by photons



How it is built

Photons are routed through interferometers etched into silicon waveguides and measured by superconducting detectors. Discrete-variable systems encode in one photon; continuous-variable systems use squeezed light. It is standard silicon photonics, so PsiQuantum builds on a GlobalFoundries line.

WHO BUILDS IT

PsiQuantum · Xanadu · Quandela · ORCA Computing · QuiX · Photonic Inc · Q.ANT

20 °C

operating temperature of the chip itself, no dilution fridge

35

photonic chips linked over 13 km of fibre by Xanadu Aurora

2027

PsiQuantum's target for its first utility-scale site

● Where it wins

- Networking
- Room-temperature photonics

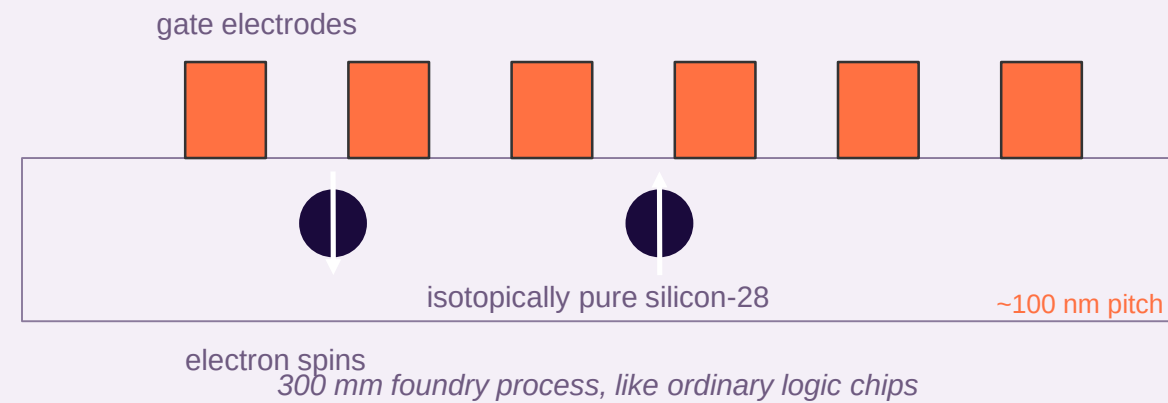
● Where it struggles

- Photon loss
- High potential but also high uncertainty

Silicon spin and quantum dot qubits

One electron is confined in a quantum dot etched into purified silicon. This is trying to use technology very similar to conventional semiconductor manufacturing.

PHYSICAL STRUCTURE



How it is built

Gates on a silicon-28 wafer trap one electron each; spin up and down are the qubit states. The footprint is roughly 100 nm, thousands of times smaller than a transmon. Several vendors now run above one kelvin, making the cryogenics far cheaper.

WHO BUILDS IT

Diraq · Silicon Quantum Computing · Quantum Motion · Intel · Equal1 · Quobly · imec (foundry)

8

qubit array demonstrated on imec's 300 mm foundry process, July 2026

99.99%

two-qubit fidelity reached by SQC in December 2025

1 K

operating temperature for hot-qubit devices, versus 15 mK

Where it wins

- Smallest footprint: the only credible route to millions on a chip
- Built on existing 300 mm CMOS lines and supply chains
- Cuts cryogenic cost sharply

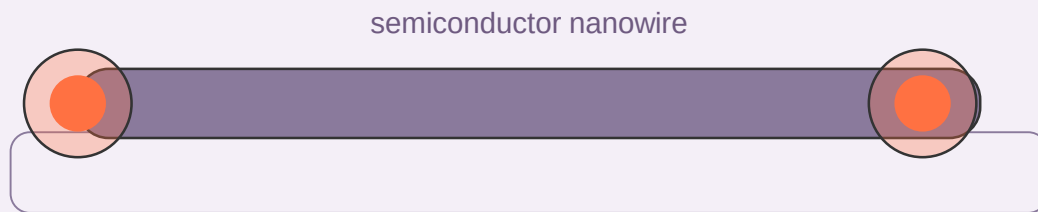
Where it struggles

- Least mature by count
- No cloud access
- Charge noise and device variability unsolved at array scale

Topological qubits

Information is stored non-locally, across a semiconductor nanowire rather than in any single particle.

PHYSICAL STRUCTURE



superconductor
one qubit stored across both ends at once

Four-qubit prototype unit cell, mid-2026

How it is built

A nanowire coupled to a superconductor hosts Majorana zero modes at each end. One qubit is stored across both modes jointly, so a local disturbance cannot corrupt it. Microsoft Majorana 2 (June 2026) reports a twenty-second mean qubit lifetime.

WHO BUILDS IT

Microsoft · (research programmes at Delft, Copenhagen and Sydney)

4

qubits in the tested Majorana 2 prototype unit cell

1,000×

reliability improvement claimed over Majorana 1

10:1

hoped-for physical-to-logical ratio, against roughly 1,000:1 today

Where it wins

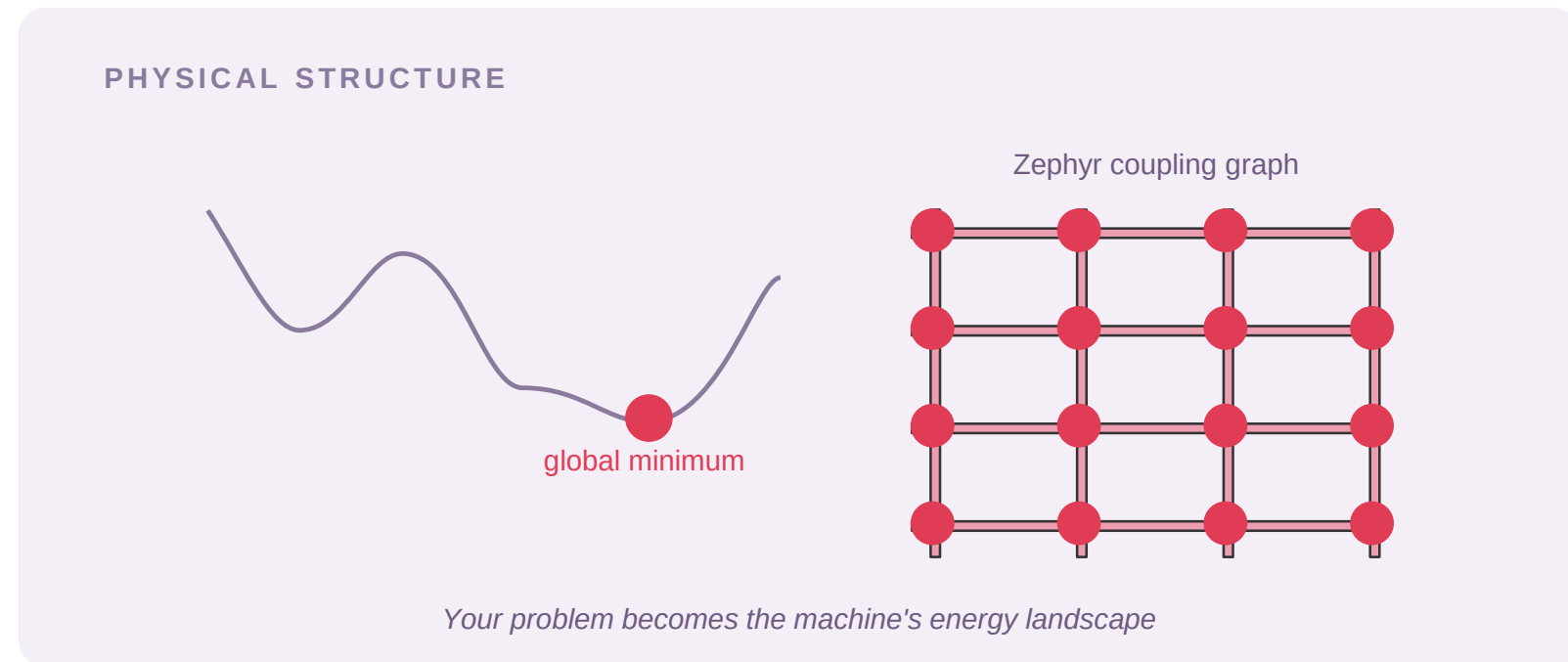
- Highest theoretical ceiling: correction built into the physics
- Targeting a million qubits on one chip

Where it struggles

- Least proven: no error-corrected computation as of mid-2026
- The physics has been contested, including a retracted paper
- A four-qubit prototype is not a machine
- Single-vendor: no fallback if Microsoft stalls

Quantum annealers

Not a universal computer. A tunable system that relaxes toward the lowest-energy configuration of a problem you encode into it. It solves optimisation, and only optimisation.



How it is built

Thousands of flux qubits are wired into a fixed graph (Zephyr). A problem is written as a QUBO, its coefficients are set into the biases and couplers, and the system anneals to the lowest-energy state, read out as the answer. Superconducting hardware, but a different computational model.

WHO BUILDS IT

D-Wave · (Fujitsu, Toshiba and NEC sell classical Ising machines that compete for the same workloads)

4,400+

qubits on Advantage2, more than any gate-model system

2M

variables handled through the hybrid solver stack

200M+

problems submitted to D-Wave systems to date

● Where it wins

- The only platform in routine production: Volkswagen & Mastercard
- Maps directly onto QUBO: portfolio, scheduling, allocation
- Mature wrapper: Leap in 40+ countries, SOC 2 Type 2

● Where it struggles

- No Shor, Grover or chemistry. Not a general-purpose computer
- Advantage over tuned classical heuristics stays contested
- Single-vendor, though Fujitsu and Toshiba compete classically

Who actually builds what & can you get access?

Systems mapped to their modality. If a vendor is not on a cloud today, it cannot be benchmarked, and that shapes the shortlist.

● Superconducting

IBM

Nighthawk 120q · Heron

Cloud

Google Quantum AI

Willow 105q

Research

Rigetti

Cepheus-1 108q

Cloud

IQM · AWS · SEEQC

On-prem, cat-qubit R&D

Mixed

● Trapped ion

Quantinuum

Helios 98q · 48 logical

Cloud/on-prem

IonQ

Forte · Tempo, #AQ 64

Cloud

AQT · eleQtron

Rack-scale, Europe

Cloud

Oxford Ionics

Now part of IonQ

Merged

● Neutral atom

QuEra

Gemini 260q · Libra 28

Braket

Pasqal

Orion, 5+ sites

Cloud/on-prem

Atom Computing

Phoenix · Magne (MSFT)

Azure

Infleqtion · planqc

Sqale, 12 logical

Limited

● Photonic

PsiQuantum

Omega, utility-scale

None

Xanadu

Aurora · Borealis

Cloud

Quandela

MosaiQ · BELENOS

Cloud

ORCA · QuiX · Q.ANT

PT-1 and similar

On-prem

● Silicon spin

Diraq

8q on imec 300 mm

Research

SQC

Watermelon platform

Research

Quantum Motion

First unit, UK NQCC

Testbed

Intel · Equal1 · Quobly

Tunnel Falls 12q

Research

● Topological

Microsoft

Majorana 2, 4q prototype

Not available

● Annealing

D-Wave

Advantage2, 4,400+q

Cloud/on-prem

D-Wave gate model

Dual-rail (QCI)

2026

Fujitsu · Toshiba

Classical Ising

Cloud

Comparing the modalities on the axes that matter

Where each platform stands in mid-2026. Maturity means commercial readiness, not scientific merit.

MODALITY	COMMERCIAL MATURITY	BEST-IN-CLASS AT	HARDEST CONSTRAINT	REALISTIC WINDOW
● Superconducting	Production pilots	Ecosystem depth and gate speed	Coherence caps circuit depth	Now
● Trapped ion	Production pilots	Fidelity, connectivity, logical qubits	Slow gates, low qubit counts	Now
● Neutral atom	Early commercial	Verified logical qubit records	Optical control at scale	2026–28
● Photonic	Early commercial	Room temperature, native networking	Photon loss; no interim product	2027–29
● Silicon spin	Research	Footprint and manufacturability	Only tens of qubits so far	2029+
● Topological	Research	Theoretical error-correction ceiling	Unproven physics, 4q prototype	2029+
● Quantum annealing	In production	QUBO optimisation, today	Not a universal computer	Now

Start from the workload, not the machine

- SC

Superconducting
- TI

Trapped ion
- NA

Neutral atom
- P

Photonic
- SS

Silicon spin
- T

Topological
- QA

Quantum annealing

Each algorithm points at a modality. Reading this table right to left is how most organisations get it wrong.

WORKLOAD	ALGORITHM FAMILY	FITS	WHY	WISER'S CAVEAT
Portfolio construction, index tracking, rebalancing	QUBO / QAOA	QA TI	Annealing today; QAOA in research	Benchmark against quantum-inspired solvers first. They often win.
Derivative pricing, VaR, credit exposure, stress testing	Amplitude Estimation	TI SC	Needs deep, high-fidelity circuits	Goldman put it at ~7,500 logical qubits for a real speedup.
Credit risk, fraud and anomaly detection	Quantum kernel / gradient	TI P	High fidelity, all-to-all	Advantage over modern classical ML is unproven and debated.
Materials, catalysts, battery and drug discovery	VQE, phase estimation	TI NA SC	Most credible near-term advantage	Not a core Vanguard workload, but advantage shows here first.
Many-body physics and quantum simulation	Analog Hamiltonian sim	NA QA	Analog mode, no gate compilation	Neutral atoms are natively suited. Their strongest claim.
Certified randomness and sampling	Gaussian boson sampling	P SC	Available commercially now	Narrow but shipping today.
Protecting data against future decryption	Post-quantum migration	—	Not a hardware purchase	Deadline-driven. NIST final 2024; US federal migration June 2026.

What has actually been published, and on what

Peer-reviewed or vendor-published, on named hardware. The standard of evidence to hold a vendor claim to.

● Algorithmic bond trading

HSBC and IBM, Sept 2025

IBM Heron, hybrid workflow

Up to 34% better prediction of whether a bond trade fills at the quoted price. First empirical evidence of value in live-adjacent trading.

● European option pricing

JPMorgan and IBM, 2024

127-qubit IBM Eagle

Amplitude estimation hit target precision with ~100x fewer samples than the CPU baseline. Error bars large; the problem was a vanilla call.

● Error-corrected logical qubits

QuEra, Harvard and MIT,
Nature, Jan 2026

448 neutral atoms

96 error-corrected logical qubits with below-threshold suppression. The strongest logical-qubit result published to date.

● Break-even error correction

Quantinuum, Nature, 2026

Helios, 98 barium ions

48 error-corrected logical qubits at 2:1 encoding, outperforming the physical qubits underneath.

● Below-threshold error correction

Google, Nature, Dec 2024

Willow, 105 qubits

Logical error rate halved as the code grew. First hard evidence the theory works in silicon.

● Production optimisation

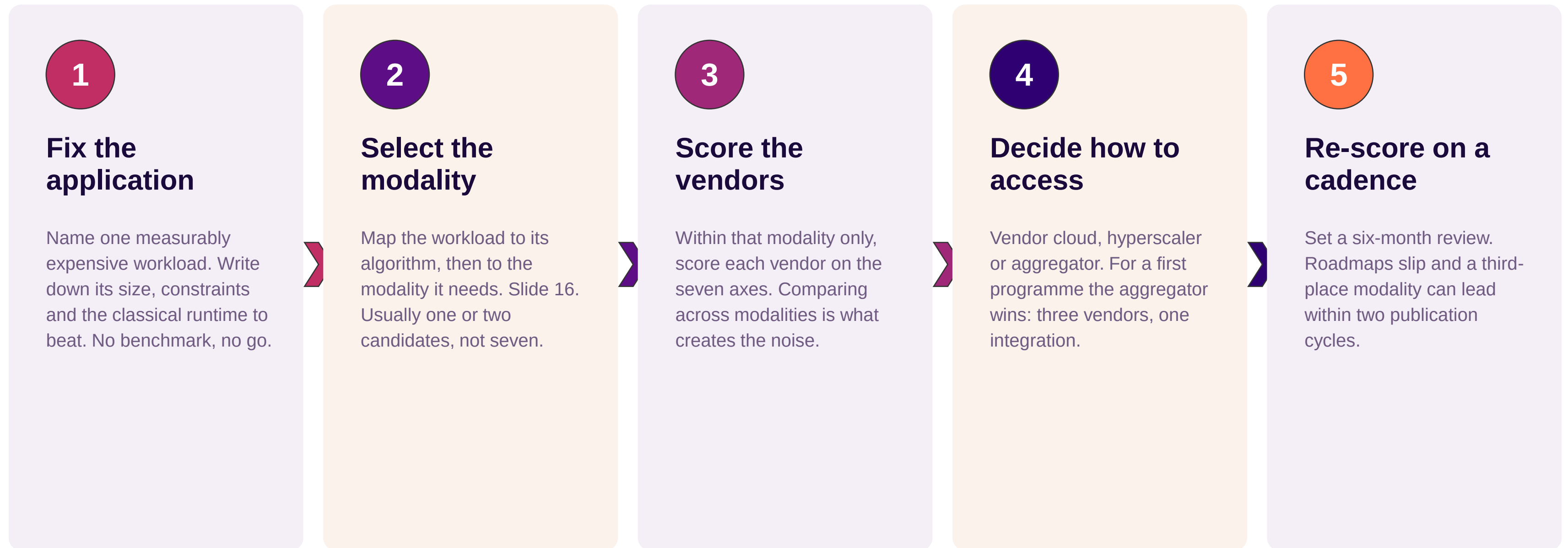
D-Wave, ongoing

Advantage2, 4,400+ qubits

200M+ problems submitted across manufacturing, logistics and finance. Commercially real, though advantage over tuned classical stays contested.

The hardware decision tree

Run in order. Skipping to step three is what makes vendor conversations unresolvable.



The output of this tree is a shortlist of two or three vendors and a benchmark plan, never a purchase order.

Step 3 in detail: the seven-axis vendor scorecard

Apply within one modality. Each axis carries a question vendors find hard to deflect.

AXIS	WHAT YOU ARE ACTUALLY MEASURING	THE QUESTION TO ASK IN THE ROOM
01 Commercial viability	Cash runway, revenue, whether the roadmap is funded	<i>“How many quarters of cash sit behind that roadmap?”</i>
02 Qubit count	Physical vs logical, error-detected vs error-corrected	<i>“How many are error-corrected, at what encoding ratio?”</i>
03 Coherence and fidelity	Median, on shipping hardware, not a prototype	<i>“Is that the median, and the system I'd be renting?”</i>
04 Platform availability	Whether a workload can run this quarter, and how	<i>“Can my team benchmark next month, at what queue depth?”</i>
05 Largest demonstration	Peer-reviewed, named hardware, classical baseline stated	<i>“Which paper, which machine, what was the baseline?”</i>
06 Cost	All-in: access, integration, staff, reformulation time	<i>“What did your last three customers spend in year one?”</i>
07 Implementation support	Architects, references, who formulates the problem	<i>“Who writes the QUBO, yours or mine?”</i>

Score each axis 1 to 5 and weight them to Vanguard's own priorities. The weighting conversation is usually more revealing than the scores.

Vendor profiles

Four headings, following the source documentation. Ask every vendor to fill the same four boxes.

IBM Superconducting · Nighthawk 120q	Quantinuum Trapped ion · Helios 98q	IonQ Trapped ion · Tempo, #AQ 64
ALGORITHMS Gate-model universal. Deepest library across optimisation, chemistry and finance via Qiskit.	ALGORITHMS Gate-model universal, strongest error-correction tooling. Guppy, Nexus, Selene.	ALGORITHMS Gate-model universal. Focus on optimisation, fraud and anomaly detection, simulation.
PLATFORM APPLICATIONS IBM Quantum Platform, Premium and Flex tiers. Also resold via Strangeworks.	PLATFORM APPLICATIONS Cloud and on-prem, NVIDIA GPUs in the control loop. Finance customers including JPMorganChase.	PLATFORM APPLICATIONS On AWS Braket, Azure and Google Cloud. The easiest trapped-ion system to reach.
COMMERCIAL MATURITY Most established programme in the sector, no financing risk. Targets quantum advantage by end of 2026.	COMMERCIAL MATURITY Honeywell-backed, publishes peer-reviewed data. Roadmap: Sol 2027, Apollo ~2030.	COMMERCIAL MATURITY Publicly listed, acquisitive, heavy losses. Targets 2 million physical qubits by 2030.
HARDWARE EVALUATION STATUS Available and easy to benchmark. No verified logical qubits yet; Starling (~200) is a 2029 target.	HARDWARE EVALUATION STATUS The strongest choice if fidelity beats raw scale. 48 error-corrected logical qubits shipping.	HARDWARE EVALUATION STATUS Easy to benchmark. Watch prototype vs shipping: 99.99% is prototype, 99.6% commercial.

Vendor profiles

Also tracking: Rigetti, IQM, Xanadu, PsiQuantum, Diraq, SQC and Microsoft.

QuEra Neutral atom · Gemini 260q	D-Wave Annealing · Advantage2, 4,400+q	Google, Pasqal and others Superconducting and neutral atom
ALGORITHMS Analog simulation plus gate-model. Strongest published error-correction work of any vendor.	ALGORITHMS QUBO optimisation only. No Shor, Grover or chemistry. Hybrid solvers reach 2M variables.	ALGORITHMS Google Willow leads error-correction physics. Pasqal runs analog and digital on one machine.
PLATFORM APPLICATIONS AWS Braket and direct access. Deep Harvard and MIT research pipeline.	PLATFORM APPLICATIONS Leap cloud in 40+ countries, SOC 2 Type 2. On-prem sold outright. Stride adds ML surrogates.	PLATFORM APPLICATIONS Google is research-only. Pasqal ships room-temperature Orion units, also on Azure.
COMMERCIAL MATURITY Private, raised \$230M+ in 2025 with Google. Libra targets 256 logical qubits in 2028.	COMMERCIAL MATURITY Most commercially mature here, an IDC Leader. Bought Quantum Circuits in 2026 for a gate-model line.	COMMERCIAL MATURITY Google sells no product. Pasqal has the widest neutral-atom on-prem footprint; plans to list late 2026.
HARDWARE EVALUATION STATUS Holds the logical-qubit record: 96 from 448 atoms. On Braket, though the wrapper is young.	HARDWARE EVALUATION STATUS Fastest route to a working optimisation result in 2026. Insist on a classical head-to-head.	HARDWARE EVALUATION STATUS Google can't be procured. Pasqal can: ~3 kW, no fridge, standard data-centre power.

The layers between Vanguard and the hardware

These layers decide how much of Vanguard's work survives a change of vendor. The best protection against picking wrong.

● Access and aggregation

One account, many machines

- qBraid: 34+ devices and 18+ SDKs in one environment
- Strangeworks: neutral aggregation, low-friction route to Qiskit Runtime
- AWS Braket, Azure, IBM Platform: hyperscaler routes, partial rosters

How Vanguard benchmarks three modalities for one integration. Start here.

● Application abstraction

Write the problem, not the circuit

- Classiq: synthesises circuits from a high-level model
- Multiverse and QC Ware: finance libraries for portfolio and risk
- Qiskit, PennyLane, Cirq, tket: the SDKs your team writes in

What keeps a workload portable when the hardware ranking changes.

● Control and error suppression

Get more from the machine you have

- Q-CTRL: pulse control, a real fidelity gain on unchanged hardware
- Riverlane: real-time decoding at hardware speed
- NVIDIA CUDA-Q: GPU-accelerated hybrid orchestration

Cheapest performance gain. Often beats waiting for the next chip.

Hardware summary chart

One page for the vendor meeting. Qubit counts are vendor figures, (as of July 2026).

VENDOR	MODALITY	APPLICATION STRENGTHS	RISKS AND WEAKNESSES	QUBIT COUNT
IBM	Superconducting	Broadest algorithm coverage; deepest developer ecosystem	Short coherence; nearest-neighbour wiring; no verified logical qubits	120 physical (Nighthawk)
Google Quantum AI	Superconducting	Error-correction physics leader; first below-threshold demo	No commercial product; research access only	105 physical (Willow)
Rigetti	Superconducting	Modular chiplet scaling; accessible pricing; full-stack	Smaller ecosystem; financing risk; fidelity trails	108 physical (Cepheus-1)
Quantinuum	Trapped ion	Highest fidelity; all-to-all; only shipping logical qubits	Slow gates; low qubit count; premium pricing	98 physical 48 logical
IonQ	Trapped ion	Widest cloud availability; strong fidelity; scale roadmap	Prototype vs commercial figures; heavy losses	~100 physical #AQ 64
QuEra	Neutral atom	Verified logical-qubit record; analog sim; programmable	Younger wrapper; atom loss; optical scaling limits	260 physical 96 logical (demo)
Pasqal	Neutral atom	Room-temperature racks; analog and digital; on-prem	Gate-model maturity trails; pre-IPO profile	1,000+ physical
PsiQuantum	Photonic	Fault-tolerant-first; foundry manufacturing; utility-scale	Nothing available to customers; no interim product	Not published
Xanadu	Photonic	Modular networked photonics; PennyLane; room temperature	Sampling-focused to date; loss-limited; early	216 modes (Borealis)
Diraq / SQC	Silicon spin	Smallest footprint; 300 mm CMOS; hot-qubit operation	Only tens of qubits; no access; 2029+ horizon	8 physical (imec array)
Microsoft	Topological	Highest theoretical ceiling; potential 10:1 encoding	Unproven physics; no error-corrected demo; single vendor	4 physical (prototype)
D-Wave	Annealing	Production optimisation today; 2M-variable hybrid solvers	Not universal; advantage contested vs classical	4,400+ physical

Logical figures are peer-reviewed demonstrations, not shipping capacity, except Quantinuum's 48. Verify every number with the vendor before it enters a business case.

Five things to take away

01

There is no best machine, only a best fit

Superconducting, trapped ion and annealing are all usable today, for different things. A vendor claiming general superiority is selling.

02

Qubit count is the least useful number

D-Wave has 45x the qubits of Quantinuum Helios and zero logical qubits. Both true. Neither settles anything.

03

Access before ownership

A neutral layer benchmarks three modalities against one workload for a fraction of a procurement cycle, and keeps the exit cheap.

04

Post-quantum crypto is urgent, and not a hardware problem

NIST standards final since 2024; US federal migration mandated June 2026. Harvest-now-decrypt-later started the clock years ago.

05

The ranking will change, so build in the review

Neutral atoms went from outsider to record-holder in eighteen months. Whatever Vanguard concludes needs a six-month expiry.

What we suggest Vanguard does next

A ninety-day plan.

Weeks 1–3

Choose the workload

Pick one problem Vanguard already pays to solve. Document its size, constraints and classical runtime. This is the benchmark.

Weeks 4–6

Stand up access

One aggregator account, not three vendor contracts. Reformulate the workload once, so it retargets without a rewrite.

Weeks 7–11

Run the head-to-head

Same problem, three modalities, plus a tuned classical baseline. Without the baseline the results mean nothing.

Week 12

Score and decide

Apply the scorecard to the vendors still standing. Publish internally with a six-month review date.

HOW WISER CAN HELP

- Facilitate the workload workshop with your quant and risk teams
- Build and run the head-to-head, independent of any vendor
- Sit in on vendor evaluations as a neutral counterparty
- Maintain the six-month re-scoring against a live ecosystem map
- Scope post-quantum migration as its own programme



**Pick the problem first.
The machine follows.**

WISER · August 2026 · thewiser.org